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Title: Heavy-tailed Bayesian nonparametric adaptation.

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This paper introduces a novel Bayesian approach for non-parametric adaptation to smoothness using heavy-tailed series priors, termed the over-smoothed heavy-tailed (OT) prior. Unlike traditional methods that rely on hyperparameter tuning or explicit model selection, this strategy achieves adaptation through the heavy tails of the prior distribution, offering a distinctively Bayesian solution. The method is shown to yield adaptive posterior contraction rates that are near-minimax optimal (up to logarithmic factors) in various settings, including Gaussian regression (for both L^2 and L^∞ -norms), linear inverse problems, and anisotropic Besov classes. Additionally, tempered posteriors extend their applicability to more general models.

A key advantage of the OT prior is its simplicity; it operates directly on basis coefficients without requiring additional hyperparameters for adaptation. While other Bayesian methods mimic frequentist techniques (e.g., sieve priors for model selection or spike-and-slab for thresholding), the OT prior leverages its tail behavior for soft, automatic adaptation. Numerical simulations support the theoretical findings, demonstrating the method's effectiveness in practice. This approach provides a robust, tuning-free alternative for non-parametric Bayesian inference.

The authors outline future research directions, including applications to deep ReLU neural networks and sparse settings (Rectified Linear Unit activation function, $\text{ReLU}(x) = \max(0, x)$). The soft selection mechanism of the OT prior could be especially useful in high-dimensional or computationally intensive scenarios where exact sparsity (e.g., spike-and-slab) is impractical. These open problems highlight the potential for extending this Bayesian framework to mod-

ern machine learning and statistical challenges.